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# The Statistical Mechanics of the Rise of Artificial Intelligence in the Field of Medicine

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## ABSTRACT

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**Keywords** First keyword · Second keyword · etc.

## 0.1 Introduction

## 0.2 Results

### Communities

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### Burt's Constraint Score

| node     | constraint | community | title                                                                                   | community_label                               |
|----------|------------|-----------|-----------------------------------------------------------------------------------------|-----------------------------------------------|
| 28419290 | 1.006944   | 15        | SVMQA: support-vector-machine-based protein single-model quality assessment.            | learning, machine, prediction, protein, using |
| 28035027 | 1.006944   | 15        | QAcon: single model quality assessment using protein structural and contact information | learning, machine, prediction, protein, using |

| node     | constraint | community | title                                                                                                                                       | community_label                                   |
|----------|------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|          |            |           | with machine learning techniques.                                                                                                           |                                                   |
| 15919260 | 1.000000   | 23        | Current status of endoscopic and robotic mitral valve surgery.                                                                              | assisted, laparoscopic, robot, robotic, surgery   |
| 29389215 | 1.000000   | 7         | Deep neural networks are more accurate than humans at detecting sexual orientation from facial images.                                      | breast, cancer, deep, learning, using             |
| 32970131 | 1.000000   | 30        | Validation of a Machine Learning Algorithm to Predict 180-Day Mortality for Outpatients With Cancer.                                        | artificial, deep, intelligence, learning, machine |
| 31446493 | 1.000000   | 11        | Evaluation of a computer-aided method for measuring the Cobb angle on chest X-rays.                                                         | chest, deep, intelligence, learning, radiographs  |
| 32337662 | 1.000000   | 7         | Classification of COVID-19 patients from chest CT images using multi-objective differential evolution-based convolutional neural networks.  | breast, cancer, deep, learning, using             |
| 13678617 | 1.000000   | 7         | Solving the XOR problem and the detection of symmetry using a single complex-valued neuron.                                                 | breast, cancer, deep, learning, using             |
| 29382633 | 1.000000   | 43        | Prediction of Incident Hypertension Within the Next Year: Prospective Study Using Statewide Electronic Health Records and Machine Learning. | artificial, deep, intelligence, learning, machine |
| 29069282 | 1.000000   | 0         | Chromatin accessibility prediction via a hybrid deep convolutional neural network.                                                          | deep, drug, learning, prediction, protein         |
| 31525737 | 1.000000   | 6         | Preoperative Prediction of Pancreatic Neuroendocrine Neoplasms Grading Based                                                                | artificial, cancer, deep, intelligence, learning  |

| node     | constraint | community | title                                                                                                                                               | community_label                                   |
|----------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|          |            |           | on Enhanced Computed Tomography Imaging: Validation of Deep Learning with a Convolutional Neural Network.                                           |                                                   |
| 27026453 | 1.000000   | 28        | Accuracy of machine learning classifiers using bilateral data from a Scheimpflug camera for identifying eyes with preclinical signs of keratoconus. | based, detection, keratoconus, learning, machine  |
| 36996195 | 1.000000   | 0         | Enzyme function prediction using contrastive learning.                                                                                              | deep, drug, learning, prediction, protein         |
| 33262328 | 1.000000   | 0         | Deep learning suggests that gene expression is encoded in all parts of a co-evolving interacting gene regulatory structure.                         | deep, drug, learning, prediction, protein         |
| 36924732 | 1.000000   | 30        | A review of deep learning-based multiple-lesion recognition from medical images: classification, detection and segmentation.                        | artificial, deep, intelligence, learning, machine |
| 10338015 | 1.000000   | 0         | Prediction of the location and type of beta-turns in proteins using neural networks.                                                                | deep, drug, learning, prediction, protein         |
| 29540156 | 1.000000   | 7         | Assessing microscope image focus quality with deep learning.                                                                                        | breast, cancer, deep, learning, using             |
| 31760193 | 1.000000   | 7         | Automatic kidney segmentation in ultrasound images using subsequent boundary distance regression and pixelwise classification networks.             | breast, cancer, deep, learning, using             |
| 31444598 | 1.000000   | 43        | Radiomics machine-learning signature for diagnosis of                                                                                               | artificial, deep, intelligence, learning, machine |

| node     | constraint | community | title                                                                                                                                                           | community_label                                  |
|----------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
|          |            |           | hepatocellular carcinoma in cirrhotic patients with indeterminate liver nodules.                                                                                |                                                  |
| 30989173 | 1.000000   | 26        | Biomedical image augmentation using Augmentor.                                                                                                                  | ct, deep, image, learning, using                 |
| 30705340 | 1.000000   | 26        | Multi-Channel 3D Deep Feature Learning for Survival Time Prediction of Brain Tumor Patients Using Multi-Modal Neuroimages.                                      | ct, deep, image, learning, using                 |
| 22195225 | 1.000000   | 29        | Automatic identification of critical follow-up recommendation sentences in radiology reports.                                                                   | language, machine, natural, processing, using    |
| 32697872 | 1.000000   | 26        | The artificial intelligence-assisted cytology diagnostic system in large-scale cervical cancer screening: A population-based cohort study of 0.7 million women. | ct, deep, image, learning, using                 |
| 19818675 | 1.000000   | 7         | A review of 3D vessel lumen segmentation techniques: models, features and extraction schemes.                                                                   | breast, cancer, deep, learning, using            |
| 23517734 | 1.000000   | 35        | Effectiveness of robot-assisted therapy on ankle rehabilitation—a systematic review.                                                                            | rehabilitation, robot, robotic, stroke, training |
| 35988335 | 1.000000   | 26        | The role of deep learning in urban water management: A critical review.                                                                                         | ct, deep, image, learning, using                 |
| 33597262 | 1.000000   | 7         | Using deep learning for dermatologist-level detection of suspicious pigmented skin lesions from wide-field images.                                              | breast, cancer, deep, learning, using            |
| 22206800 | 1.000000   | 23        | Perioperative outcomes of robot-assisted radical                                                                                                                | assisted, laparoscopic, robot, robotic, surgery  |

| node     | constraint | community | title                                                                                                                                                                          | community_label                                     |
|----------|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|          |            |           | prostatectomy compared with open radical prostatectomy: results from the nationwide inpatient sample.                                                                          |                                                     |
| 32090528 | 1.000000   | 26        | Low-Dose Abdominal CT Using a Deep Learning-Based Denoising Algorithm: A Comparison with CT Reconstructed with Filtered Back Projection or Iterative Reconstruction Algorithm. | ct, deep, image, learning, using                    |
| 30980127 | 1.000000   | 13        | Head and neck squamous cell carcinoma: prediction of cervical lymph node metastasis by dual-energy CT texture analysis with machine learning.                                  | artificial, deep, detection, intelligence, learning |
| 33588711 | 1.000000   | 15        | An Experimental Review on Deep Learning Architectures for Time Series Forecasting.                                                                                             | learning, machine, prediction, protein, using       |
| 33586361 | 1.000000   | 30        | Radiographers' perspectives on the emerging integration of artificial intelligence into diagnostic imaging: The Ghana study.                                                   | artificial, deep, intelligence, learning, machine   |
| 29698607 | 1.000000   | 0         | Most Ligand-Based Classification Benchmarks Reward Memorization Rather than Generalization.                                                                                    | deep, drug, learning, prediction, protein           |
| 28057016 | 1.000000   | 35        | The effectiveness of robotic-assisted gait training for paediatric gait disorders: systematic review.                                                                          | rehabilitation, robot, robotic, stroke, training    |
| 30607868 | 1.000000   | 6         | Machine learning-based radiomic models to predict intensity-modulated radiation therapy response,                                                                              | artificial, cancer, deep, intelligence, learning    |

| node     | constraint | community | title                                                                                                                                                                    | community_label                                     |
|----------|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|          |            |           | Gleason score and stage in prostate cancer.                                                                                                                              |                                                     |
| 16986543 | 1.000000   | 0         | Rotation forest: A new classifier ensemble method.                                                                                                                       | deep, drug, learning, prediction, protein           |
| 34791021 | 1.000000   | 13        | Machine learning meets omics: applications and perspectives.                                                                                                             | artificial, deep, detection, intelligence, learning |
| 32330208 | 1.000000   | 7         | Machine learning using intrinsic genomic signatures for rapid classification of novel pathogens: COVID-19 case study.                                                    | breast, cancer, deep, learning, using               |
| 29704196 | 1.000000   | 13        | Exploiting the potential of unlabeled endoscopic video data with self-supervised learning.                                                                               | artificial, deep, detection, intelligence, learning |
| 31230076 | 1.000000   | 26        | Diagnostic accuracy of 3D deep-learning-based fully automated estimation of patient-level minimum fractional flow reserve from coronary computed tomography angiography. | ct, deep, image, learning, using                    |
| 28901583 | 1.000000   | 15        | Improved protein contact predictions with the MetaPSICOV2 server in CASP12.                                                                                              | learning, machine, prediction, protein, using       |
| 23845911 | 1.000000   | 29        | Semi-supervised clinical text classification with Laplacian SVMs: an application to cancer case management.                                                              | language, machine, natural, processing, using       |
| 19968873 | 1.000000   | 35        | Performance adaptive training control strategy for recovering wrist movements in stroke patients: a preliminary, feasibility study.                                      | rehabilitation, robot, robotic, stroke, training    |
| 16284202 | 1.000000   | 33        | Kernel-based machine learning protocol for predicting DNA-binding proteins.                                                                                              | machine, recognition, support, using, vector        |

| node     | constraint | community | title                                                                                                                                                                                 | community_label                                     |
|----------|------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 15732387 | 1.000000   | 5         | Smooth function approximation using neural networks.                                                                                                                                  | artificial, network, networks, neural, using        |
| 29074582 | 1.000000   | 20        | A generative vision model that trains with high data efficiency and breaks text-based CAPTCHAs.                                                                                       | deep, learning, networks, neural, using             |
| 18621580 | 1.000000   | 21        | A brain-actuated wheelchair: asynchronous and non-invasive Brain-computer interfaces for continuous control of robots.                                                                | based, classification, deep, learning, neural       |
| 30975611 | 1.000000   | 13        | Artificial Intelligence May Cause a Significant Disruption to the Radiology Workforce.                                                                                                | artificial, deep, detection, intelligence, learning |
| 32339081 | 1.000000   | 11        | Artificial Intelligence Augmentation of Radiologist Performance in Distinguishing COVID-19 from Pneumonia of Other Origin at Chest CT.                                                | chest, deep, intelligence, learning, radiographs    |
| 28419261 | 1.000000   | 29        | Challenges in adapting existing clinical natural language processing systems to multiple, diverse health care settings.                                                               | language, machine, natural, processing, using       |
| 22627647 | 1.000000   | 29        | Automated identification of patients with pulmonary nodules in an integrated health system using administrative health plan data, radiology reports, and natural language processing. | language, machine, natural, processing, using       |
| 19130975 | 1.000000   | 23        | Transoral robotic assisted free flap reconstruction.                                                                                                                                  | assisted, laparoscopic, robot, robotic, surgery     |
| 29395652 | 1.000000   | 20        | Sigmoid-weighted linear units for neural network function approximation in reinforcement learning.                                                                                    | deep, learning, networks, neural, using             |

| node     | constraint | community | title                                                                                                                                           | community_label                                     |
|----------|------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 32971995 | 1.000000   | 11        | Unveiling COVID-19 from CHEST X-Ray with Deep Learning: A Hurdles Race with Small Data.                                                         | chest, deep, intelligence, learning, radiographs    |
| 31675322 | 1.000000   | 7         | A 3D Probabilistic Deep Learning System for Detection and Diagnosis of Lung Cancer Using Low-Dose CT Scans.                                     | breast, cancer, deep, learning, using               |
| 32324588 | 1.000000   | 15        | Automatically Designing CNN Architectures Using the Genetic Algorithm for Image Classification.                                                 | learning, machine, prediction, protein, using       |
| 28930562 | 1.000000   | 21        | Alzheimer's disease diagnostics by a 3D deeply supervised adaptable convolutional network.                                                      | based, classification, deep, learning, neural       |
| 31150344 | 1.000000   | 0         | DeepDSC: A Deep Learning Method to Predict Drug Sensitivity of Cancer Cell Lines.                                                               | deep, drug, learning, prediction, protein           |
| 25452135 | 1.000000   | 0         | Discrimination of acidic and alkaline enzyme using Chou's pseudo amino acid composition in conjunction with probabilistic neural network model. | deep, drug, learning, prediction, protein           |
| 16439182 | 1.000000   | 32        | SPASM: a 3D-ASM for segmentation of sparse and arbitrarily oriented cardiac MRI data.                                                           | based, biomedical, deep, learning, machine          |
| 18222214 | 1.000000   | 23        | Video-assisted thoracoscopic surgery versus robotic-assisted thoracoscopic surgery thymectomy.                                                  | assisted, laparoscopic, robot, robotic, surgery     |
| 30074930 | 1.000000   | 13        | Supervised Machine-learning Predictive Analytics for Prediction of Postinduction Hypotension.                                                   | artificial, deep, detection, intelligence, learning |



| node     | constraint | community | title                                                                                                                                                                                                                | community_label                                     |
|----------|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 23518764 | 1.000000   | 23        | Mobility experiments with microrobots for minimally invasive intraocular surgery.                                                                                                                                    | assisted, laparoscopic, robot, robotic, surgery     |
| 33589635 | 1.000000   | 20        | Fast and precise single-cell data analysis using a hierarchical autoencoder.                                                                                                                                         | deep, learning, networks, neural, using             |
| 33141003 | 1.000000   | 13        | Deep Learning for Detecting Cerebral Aneurysms with CT Angiography.                                                                                                                                                  | artificial, deep, detection, intelligence, learning |
| 32763853 | 1.000000   | 11        | Fine-Tuning U-Net for Ultrasound Image Segmentation: Different Layers, Different Outcomes.                                                                                                                           | chest, deep, intelligence, learning, radiographs    |
| 32024849 | 1.000000   | 6         | A deep learning system accurately classifies primary and metastatic cancers using passenger mutation patterns.                                                                                                       | artificial, cancer, deep, intelligence, learning    |
| 32696256 | 1.000000   | 43        | Preoperative prediction for pathological grade of hepatocellular carcinoma via machine learning-based radiomics.                                                                                                     | artificial, deep, intelligence, learning, machine   |
| 29079522 | 1.000000   | 26        | Image processing and Quality Control for the first 10,000 brain imaging datasets from UK Biobank.                                                                                                                    | ct, deep, image, learning, using                    |
| 25850089 | 1.000000   | 35        | Effects of Innovative WALKBOT Robotic-Assisted Locomotor Training on Balance and Gait Recovery in Hemiparetic Stroke: A Prospective, Randomized, Experimenter Blinded Case Control Study With a Four-Week Follow-Up. | rehabilitation, robot, robotic, stroke, training    |
| 30890116 | 1.000000   | 30        | Machine Learning-Based Model for Prediction of Outcomes in Acute Stroke.                                                                                                                                             | artificial, deep, intelligence, learning, machine   |

| node     | constraint | community | title                                                                                                                  | community_label                                     |
|----------|------------|-----------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 30423072 | 1.000000   | 0         | Off-target predictions in CRISPR-Cas9 gene editing using deep learning.                                                | deep, drug, learning, prediction, protein           |
| 22868357 | 1.000000   | 23        | Robot-assisted minimally invasive distal pancreatectomy is superior to the laparoscopic technique.                     | assisted, laparoscopic, robot, robotic, surgery     |
| 20153886 | 1.000000   | 23        | Survival outcomes for women undergoing type III robotic radical hysterectomy for cervical cancer: a 3-year experience. | assisted, laparoscopic, robot, robotic, surgery     |
| 28892672 | 1.000000   | 21        | Spiking neural P systems with multiple channels.                                                                       | based, classification, deep, learning, neural       |
| 32767102 | 1.000000   | 43        | Simulated clinical deployment of fully automatic deep learning for clinical prostate MRI assessment.                   | artificial, deep, intelligence, learning, machine   |
| 32706386 | 1.000000   | 30        | Clinical, Legal, and Ethical Aspects of Artificial Intelligence-Assisted Conversational Agents in Health Care.         | artificial, deep, intelligence, learning, machine   |
| 20646744 | 1.000000   | 9         | Robotic versus laparoscopic partial nephrectomy: single-surgeon matched cohort study of 150 patients.                  | assisted, nephrectomy, partial, robot, robotic      |
| 12422124 | 1.000000   | 23        | Robot-assisted laparoscopic aortobifemoral bypass for aortoiliac occlusive disease: a report of two cases.             | assisted, laparoscopic, robot, robotic, surgery     |
| 32091180 | 1.000000   | 7         | Label-Free Leukemia Monitoring by Computer Vision.                                                                     | breast, cancer, deep, learning, using               |
| 33661340 | 1.000000   | 13        | Accurate age classification using manual method and deep convolutional neural network based                            | artificial, deep, detection, intelligence, learning |

| node     | constraint | community | title                                                                                                                                                                                      | community_label                                   |
|----------|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|          |            |           | on orthopantomogram images.                                                                                                                                                                |                                                   |
| 7989989  | 1.000000   | 5         | Automated interpretation of planar thallium-201-dipyridamole stress-redistribution scintigrams using artificial neural networks.                                                           | artificial, network, networks, neural, using      |
| 15565778 | 1.000000   | 33        | The pre-image problem in kernel methods.                                                                                                                                                   | machine, recognition, support, using, vector      |
| 28039166 | 1.000000   | 33        | An ensemble approach to protein fold classification by integration of template-based assignment and support vector machine classifier.                                                     | machine, recognition, support, using, vector      |
| 32013925 | 1.000000   | 43        | Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone.                                                                    | artificial, deep, intelligence, learning, machine |
| 14555619 | 1.000000   | 0         | Learning to predict protein-protein interactions from protein sequences.                                                                                                                   | deep, drug, learning, prediction, protein         |
| 32012001 | 1.000000   | 7         | Unpaired Multi-Modal Segmentation via Knowledge Distillation.                                                                                                                              | breast, cancer, deep, learning, using             |
| 19616830 | 1.000000   | 23        | Extrafascial versus interfascial nerve-sparing technique for robotic-assisted laparoscopic prostatectomy: comparison of functional outcomes and positive surgical margins characteristics. | assisted, laparoscopic, robot, robotic, surgery   |
| 31316157 | 1.000000   | 0         | Comparative performances of machine learning methods for classifying Crohn Disease patients using genome-wide genotyping data.                                                             | deep, drug, learning, prediction, protein         |

| node     | constraint | community | title                                                                                                                                        | community_label                                     |
|----------|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 17694871 | 1.000000   | 23        | Automatic brachytherapy seed placement under MRI guidance.                                                                                   | assisted, laparoscopic, robot, robotic, surgery     |
| 7605063  | 1.000000   | 20        | Neuromorphic analogue VLSI.                                                                                                                  | deep, learning, networks, neural, using             |
| 37220190 | 1.000000   | 13        | Can Artificial Intelligence Pass the American Board of Orthopaedic Surgery Examination? Orthopaedic Residents Versus ChatGPT.                | artificial, deep, detection, intelligence, learning |
| 28903538 | 1.000000   | 15        | POSSUM: a bioinformatics toolkit for generating numerical sequence feature descriptors based on PSSM profiles.                               | learning, machine, prediction, protein, using       |
| 31153553 | 1.000000   | 26        | A deep learning radiomics model for preoperative grading in meningioma.                                                                      | ct, deep, image, learning, using                    |
| 23872518 | 1.000000   | 7         | Annotating temporal information in clinical narratives.                                                                                      | breast, cancer, deep, learning, using               |
| 33263726 | 1.000000   | 5         | Multimodal Machine Learning Workflows for Prediction of Psychosis in Patients With Clinical High-Risk Syndromes and Recent-Onset Depression. | artificial, network, networks, neural, using        |
| 15572470 | 1.000000   | 7         | Optimal number of features as a function of sample size for various classification rules.                                                    | breast, cancer, deep, learning, using               |
| 25119239 | 1.000000   | 7         | Construction the model on the breast cancer survival analysis use support vector machine, logistic regression and decision tree.             | breast, cancer, deep, learning, using               |
| 32406296 | 1.000000   | 43        | Automatic Triage of 12-Lead ECGs Using Deep Convolutional Neural Networks.                                                                   | artificial, deep, intelligence, learning, machine   |
| 29079248 | 1.000000   | 13        | Machine Learning for Predicting Patient Wait                                                                                                 | artificial, deep, detection, intelligence, learning |

| node     | constraint | community | title                                                                                                                         | community_label                                     |
|----------|------------|-----------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|          |            |           | Times and Appointment Delays.                                                                                                 |                                                     |
| 19110277 | 1.000000   | 9         | Robot assisted laparoscopic partial nephrectomy: a viable and safe option in children.                                        | assisted, nephrectomy, partial, robot, robotic      |
| 37649385 | 1.000000   | 15        | Sequence-based prediction model of protein crystallization propensity using machine learning and two-level feature selection. | learning, machine, prediction, protein, using       |
| 15732389 | 1.000000   | 5         | A generalized growing and pruning RBF (GGAP-RBF) neural network for function approximation.                                   | artificial, network, networks, neural, using        |
| 35398119 | 1.000000   | 5         | Interpretation of ensemble learning to predict water quality using explainable artificial intelligence.                       | artificial, network, networks, neural, using        |
| 38281112 | 1.000000   | 32        | Improving large language models for clinical named entity recognition via prompt engineering.                                 | based, biomedical, deep, learning, machine          |
| 30703141 | 1.000000   | 26        | Automatic real-time gait event detection in children using deep neural networks.                                              | ct, deep, image, learning, using                    |
| 18003645 | 1.000000   | 29        | Kernel approaches for genic interaction extraction.                                                                           | language, machine, natural, processing, using       |
| 28921876 | 1.000000   | 43        | Validation of artificial neural networks as a methodology for donor-recipient matching for liver transplantation.             | artificial, deep, intelligence, learning, machine   |
| 32245989 | 1.000000   | 13        | Deep Learning Method for Mandibular Canal Segmentation in Dental Cone Beam Computed Tomography Volumes.                       | artificial, deep, detection, intelligence, learning |

| node     | constraint | community | title                                                                                                                                                | community_label                                   |
|----------|------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 22044973 | 1.000000   | 23        | Learning curve for robot-assisted Roux-en-Y gastric bypass.                                                                                          | assisted, laparoscopic, robot, robotic, surgery   |
| 30602393 | 1.000000   | 32        | Segmentation of lung parenchyma in CT images using CNN trained with the clustering algorithm generated dataset.                                      | based, biomedical, deep, learning, machine        |
| 32479180 | 1.000000   | 21        | Reporting and Implementing Interventions Involving Machine Learning and Artificial Intelligence.                                                     | based, classification, deep, learning, neural     |
| 28419223 | 1.000000   | 0         | Removal of batch effects using distribution-matching residual networks.                                                                              | deep, drug, learning, prediction, protein         |
| 30979733 | 1.000000   | 7         | Application of Artificial Intelligence for Preoperative Diagnostic and Prognostic Prediction in Epithelial Ovarian Cancer Based on Blood Biomarkers. | breast, cancer, deep, learning, using             |
| 33975088 | 1.000000   | 33        | Predicting the performance of anaerobic digestion using machine learning algorithms and genomic data.                                                | machine, recognition, support, using, vector      |
| 30907586 | 1.000000   | 0         | Advancing Computational Toxicology in the Big Data Era by Artificial Intelligence: Data-Driven and Mechanism-Driven Modeling for Chemical Toxicity.  | deep, drug, learning, prediction, protein         |
| 31825503 | 1.000000   | 30        | Development and Validation of a Machine Learning Model to Aid Discharge Processes for Inpatient Surgical Care.                                       | artificial, deep, intelligence, learning, machine |
| 22859645 | 1.000000   | 32        | Large-scale evaluation of automated clinical note                                                                                                    | based, biomedical, deep, learning, machine        |

| node     | constraint | community | title                                                                                                                                                                            | community_label                                   |
|----------|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|          |            |           | de-identification and its impact on information extraction.                                                                                                                      |                                                   |
| 33328125 | 1.000000   | 7         | Multiclass semantic segmentation and quantification of traumatic brain injury lesions on head CT using deep learning: an algorithm development and multicentre validation study. | breast, cancer, deep, learning, using             |
| 33264639 | 1.000000   | 26        | Deep learning prediction of proton and photon dose distributions for paediatric abdominal tumours.                                                                               | ct, deep, image, learning, using                  |
| 26833816 | 1.000000   | 0         | Sequence-based prediction of protein-peptide binding sites using support vector machine.                                                                                         | deep, drug, learning, prediction, protein         |
| 31671333 | 1.000000   | 26        | MFP-Unet: A novel deep learning based approach for left ventricle segmentation in echocardiography.                                                                              | ct, deep, image, learning, using                  |
| 29860461 | 1.000000   | 30        | Retinal Nerve Fiber Layer Features Identified by Unsupervised Machine Learning on Optical Coherence Tomography Scans Predict Glaucoma Progression.                               | artificial, deep, intelligence, learning, machine |
| 31060901 | 1.000000   | 43        | Prediction of hematoma expansion in spontaneous intracerebral hemorrhage using support vector machine.                                                                           | artificial, deep, intelligence, learning, machine |
| 31598630 | 1.000000   | 0         | Learning from the ligand: using ligand-based features to improve binding affinity prediction.                                                                                    | deep, drug, learning, prediction, protein         |

| node     | constraint | community | title                                                                                                                                                          | community_label                                     |
|----------|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 28055894 | 1.000000   | 0         | MiRTDL: A Deep Learning Approach for miRNA Target Prediction.                                                                                                  | deep, drug, learning, prediction, protein           |
| 33783247 | 1.000000   | 7         | Multiclass CBCT Image Segmentation for Orthodontics with Deep Learning.                                                                                        | breast, cancer, deep, learning, using               |
| 31443858 | 1.000000   | 5         | A new machine learning technique for an accurate diagnosis of coronary artery disease.                                                                         | artificial, network, networks, neural, using        |
| 24125142 | 1.000000   | 29        | Automated chart review for asthma cohort identification using natural language processing: an exploratory study.                                               | language, machine, natural, processing, using       |
| 37038364 | 1.000000   | 13        | Deep-Learning-Based Detection of Vertebral Fracture and Osteoporosis Using Lateral Spine X-Ray Radiography.                                                    | artificial, deep, detection, intelligence, learning |
| 15732405 | 1.000000   | 5         | Bio-basis function neural network for prediction of protease cleavage sites in proteins.                                                                       | artificial, network, networks, neural, using        |
| 30321231 | 1.000000   | 8         | Validation of deep-learning-based triage and acuity score using a large national dataset.                                                                      | artificial, learning, machine, networks, neural     |
| 31142754 | 1.000000   | 32        | Intelligent ICU for Autonomous Patient Monitoring Using Pervasive Sensing and Deep Learning.                                                                   | based, biomedical, deep, learning, machine          |
| 14759816 | 1.000000   | 29        | Exploring semantic groups through visual approaches.                                                                                                           | language, machine, natural, processing, using       |
| 27262365 | 1.000000   | 30        | An international observational study suggests that artificial intelligence for clinical decision support optimizes anemia management in hemodialysis patients. | artificial, deep, intelligence, learning, machine   |



| node     | constraint | community | title                                                                                                                                                     | community_label                                     |
|----------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 30075707 | 1.000000   | 0         | Prediction of 8-state protein secondary structures by a novel deep learning architecture.                                                                 | deep, drug, learning, prediction, protein           |
| 18483450 | 1.000000   | 21        | Control of a humanoid robot by a noninvasive brain-computer interface in humans.                                                                          | based, classification, deep, learning, neural       |
| 31909548 | 1.000000   | 7         | Using deep-learning algorithms to classify fetal brain ultrasound images as normal or abnormal.                                                           | breast, cancer, deep, learning, using               |
| 26821132 | 1.000000   | 0         | DrugMiner: comparative analysis of machine learning algorithms for prediction of potential druggable proteins.                                            | deep, drug, learning, prediction, protein           |
| 20647054 | 1.000000   | 29        | Natural Language Processing methods and systems for biomedical ontology learning.                                                                         | language, machine, natural, processing, using       |
| 34272911 | 1.000000   | 13        | The role of artificial intelligence in enhancing clinical nursing care: A scoping review.                                                                 | artificial, deep, detection, intelligence, learning |
| 11989844 | 1.000000   | 26        | A modified fuzzy C-means algorithm for bias field estimation and segmentation of MRI data.                                                                | ct, deep, image, learning, using                    |
| 35027326 | 1.000000   | 5         | Artificial intelligence in the embryology laboratory: a review.                                                                                           | artificial, network, networks, neural, using        |
| 30425279 | 1.000000   | 0         | A machine learning approach for reliable prediction of amino acid interactions and its application in the directed evolution of enantioselective enzymes. | deep, drug, learning, prediction, protein           |
| 21296482 | 1.000000   | 23        | Pentafecta: a new concept for reporting outcomes of robot-assisted laparoscopic radical prostatectomy.                                                    | assisted, laparoscopic, robot, robotic, surgery     |

| node     | constraint | community | title                                                                                                                                                              | community_label                                   |
|----------|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 32325045 | 1.000000   | 2         | Artificial Intelligence: The Future for Diabetes Care.                                                                                                             | deep, networks, retinal, segmentation, vessel     |
| 20167583 | 1.000000   | 23        | Influence of steep Trendelenburg position and CO(2) pneumoperitoneum on cardiovascular, cerebrovascular, and respiratory homeostasis during robotic prostatectomy. | assisted, laparoscopic, robot, robotic, surgery   |
| 30072292 | 1.000000   | 7         | Convolutional Neural Network Based Breast Cancer Risk Stratification Using a Mammographic Dataset.                                                                 | breast, cancer, deep, learning, using             |
| 31149787 | 1.000000   | 30        | Promising Artificial Intelligence-Machine Learning-Deep Learning Algorithms in Ophthalmology.                                                                      | artificial, deep, intelligence, learning, machine |
| 34327990 | 1.000000   | 15        | xDeep-AcPEP: Deep Learning Method for Anticancer Peptide Activity Prediction Based on Convolutional Neural Network and Multitask Learning.                         | learning, machine, prediction, protein, using     |
| 21328664 | 1.000000   | 20        | Soft robotics for chemists.                                                                                                                                        | deep, learning, networks, neural, using           |
| 15732393 | 1.000000   | 33        | Incremental training of support vector machines.                                                                                                                   | machine, recognition, support, using, vector      |
| 30423076 | 1.000000   | 0         | DeepDiff: DEEP-learning for predicting DIFFerential gene expression from histone modifications.                                                                    | deep, drug, learning, prediction, protein         |
| 33727623 | 1.000000   | 7         | Deep learning classification of lung cancer histology using CT images.                                                                                             | breast, cancer, deep, learning, using             |
| 29857330 | 1.000000   | 7         | $\Omega$ -Net (Omega-Net): Fully automatic, multi-view cardiac MR detection, orientation, and                                                                      | breast, cancer, deep, learning, using             |

| node     | constraint | community | title                                                                                                                                    | community_label                                     |
|----------|------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|          |            |           | segmentation with deep neural networks.                                                                                                  |                                                     |
| 1990429  | 1.000000   | 5         | Identification of the <sup>1</sup> H-NMR spectra of complex oligosaccharides with artificial neural networks.                            | artificial, network, networks, neural, using        |
| 1289469  | 1.000000   | 5         | Using an artificial neural network to diagnose hepatic masses.                                                                           | artificial, network, networks, neural, using        |
| 23682001 | 1.000000   | 6         | Toward open set recognition.                                                                                                             | artificial, cancer, deep, intelligence, learning    |
| 15901399 | 1.000000   | 37        | A unified approach to building and controlling spiking attractor networks.                                                               | integration, large, model, neurons, spiking         |
| 30334750 | 1.000000   | 13        | Deep Geodesic Learning for Segmentation and Anatomical Landmarking.                                                                      | artificial, deep, detection, intelligence, learning |
| 32181729 | 1.000000   | 7         | Computer-aided Detection of Brain Metastases in T1-weighted MRI for Stereotactic Radiosurgery Using Deep Learning Single-Shot Detectors. | breast, cancer, deep, learning, using               |
| 30990425 | 1.000000   | 0         | Approximating the Ideal Observer and Hotelling Observer for Binary Signal Detection Tasks by Use of Supervised Learning Methods.         | deep, drug, learning, prediction, protein           |
| 19806602 | 1.000000   | 23        | Robotic-assisted single-incision right colectomy: early experience.                                                                      | assisted, laparoscopic, robot, robotic, surgery     |
| 31602691 | 1.000000   | 33        | Unsupervised machine learning for exploratory data analysis in imaging mass spectrometry.                                                | machine, recognition, support, using, vector        |
| 19258690 | 1.000000   | 20        | Design of a biomimetic robotic octopus arm.                                                                                              | deep, learning, networks, neural, using             |
| 10326852 | 1.000000   | 23        | Self-guided robotic camera control for laparoscopic surgery compared with human camera control.                                          | assisted, laparoscopic, robot, robotic, surgery     |

| node     | constraint | community | title                                                                                                                       | community_label                                     |
|----------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 31151025 | 1.000000   | 15        | THPep: A machine learning-based approach for predicting tumor homing peptides.                                              | learning, machine, prediction, protein, using       |
| 35138133 | 1.000000   | 13        | Is Artificial Intelligence (AI) a Pipe Dream? Why Legal Issues Present Significant Hurdles to AI Autonomy.                  | artificial, deep, detection, intelligence, learning |
| 30616095 | 1.000000   | 0         | A comparison of deep networks with ReLU activation function and linear spline-type methods.                                 | deep, drug, learning, prediction, protein           |
| 35523534 | 1.000000   | 30        | New meaning for NLP: the trials and tribulations of natural language processing with GPT-3 in ophthalmology.                | artificial, deep, intelligence, learning, machine   |
| 30598075 | 1.000000   | 0         | Convolutional neural network based on SMILES representation of compounds for detecting chemical motif.                      | deep, drug, learning, prediction, protein           |
| 37338353 | 1.000000   | 7         | Combining Deep Learning and Handcrafted Radiomics for Classification of Suspicious Lesions on Contrast-enhanced Mammograms. | breast, cancer, deep, learning, using               |

## 0.3 Discussion

## 0.4 Methodology

### 0.4.1 Descriptive Characteristics

#### Artificial Intelligence

Number of articles in ARTIFICIAL\_INTELLIGENCE: 153802

#### Gene Expression

We started with 163,977 articles in the gene expression category.

After filtering for nodes above the k-elbow point, our dataset representing artificial intelligence articles composes 1,822 articles, represented as nodes. After filtering for nodes above the k-elbow

point, our dataset representing gene expression articles composes 1,558 articles, represented as nodes